



Conclusion & Summary

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Outline of Today

Time	Topics
14:00	Introduction & Welcome
	Surrogate models vs. digital twins
	Types & Implementations of surrogates
14:15	Recap of Machine Learning (ML) and Deep Learning
	Basic concepts, evaluation, hyper-parameter tuning, etc.
	Surrogate models: Challenges for ML
	Data-driven mitigation strategies (i.e., scenario generation)
14:45	Physics-informed/inspired Methods -- Part I
	Intro to Physics-informed Neural Networks (PINNs)
	Hands-On: How to build and train a PINN in Python
15:30	Coffee Break
16:00	Physics-informed/inspired Methods -- Part II
	Operators vs. PINNs
16:25	Case study: SOTA Hydraulic Surrogate Model
	Hydraulic surrogates and their applications
	Analyzing the problem and identifying relevant quantities and laws
	A SOTA Hydraulic Surrogate
	Hands-On: Usage and Evaluation of the Surrogate in Python
17:15	Closing, Summary, and Q&A
17:30	END

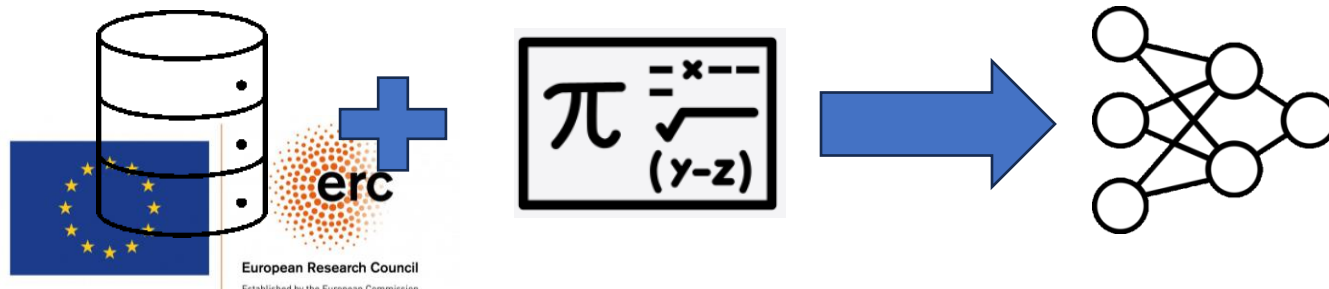
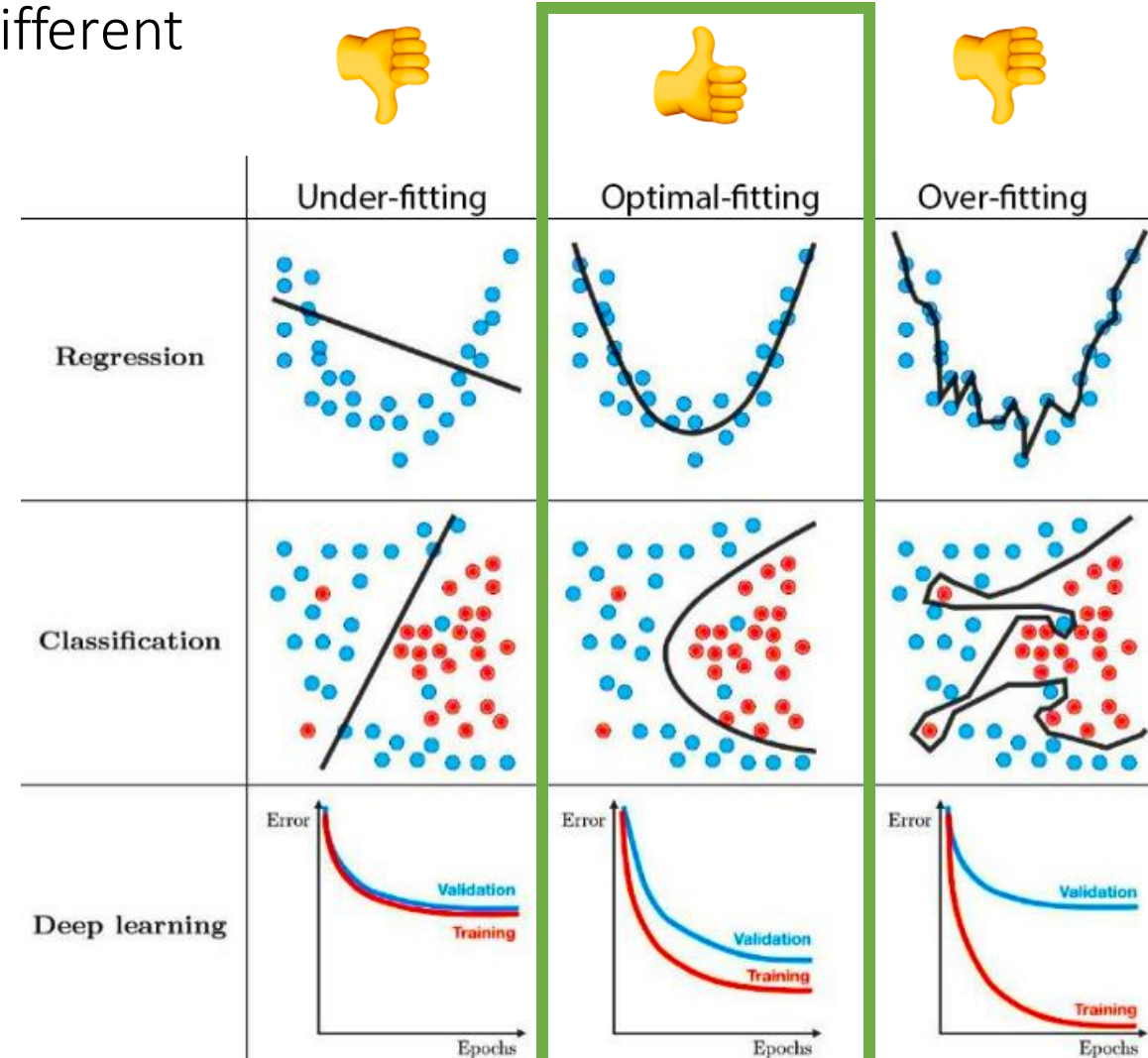
We are here!



Machine Learning for Surrogate Models

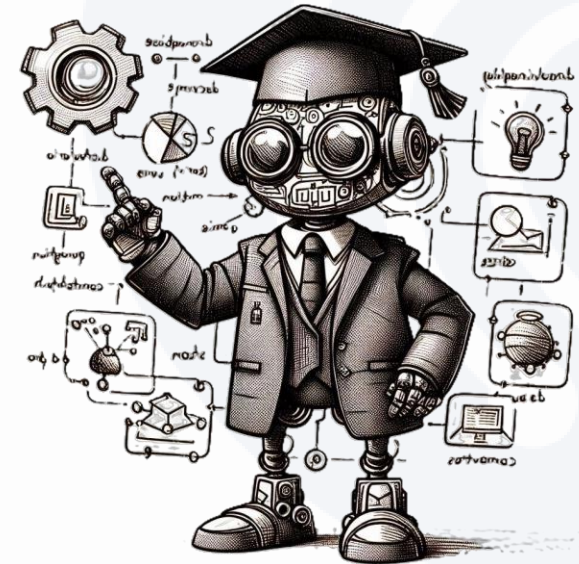
- Many different neural network architectures
 - Different domains/applications require different architectures
- **Challenges**
 - Non-causal correlations & biases
 - Uncertainties
- **Solutions**
 - Regularization
 - More & better data
 - Inductive bias => *Physics-informed ML*

A mostly complete chart of
Neural Networks



PINNs and Neural Operators

- PINNs
 - Learn the solution function of a PDE
 - For a single Setup of a single PDE
 - Use the PDE derivative structure as a loss
 - Don't require training data
 - The rabbit hole extends:
E.g. Inverse Problems
- Neural Operators
 - Generalize PINNs: Learn the Solution Operator
 - Universal Function Approximation extends to Operators
 - Neural Operator Idea useful for Surrogates
 - Generalization to many PDE Setups (ICs, BCs, PDE Parameters)
 - Typically Surrogates are Operators



Case Study – A Physics-Informed Hydraulic Surrogate Model

- Hydraulic simulators are computationally expensive and non-differentiable.
- We present a Physics-Informed GNN based surrogate model that is
 - Faster,
 - Generalizable,
 - And differentiable.
- The proposed surrogate model can be used as a digital twin under certain assumptions for tasks like
 - Diameter optimization,
 - Sparse-to-dense state estimation.
- Physics-Informed learning ensures adherence to physical laws but can present challenges as
 - Data normalization,
 - And exploding gradients.



Q&A

Any questions?

